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Sowing the machine

*Artificial intelligence and the future of agriculture: promise, power, and the terms of cultivation***Rapporteur: Kem Laurin-Lubin****Attendees:** Matthew S.W. Silk · Samantha St. Amand · Kem-Laurin Lubin · Patrick Matlock · Arnim Kangaloo · Shani Biniaminov · Malik Mujtaba · Noah Robles · Kavitha Ramachandran*This paper emerged from the Institute's interdisciplinary roundtable series: knowledge made in conversation, and verified in research.*

About this paper

This paper is a knowledge asset of the *Ethics of AI Institute*, an independent interdisciplinary organization dedicated to advancing understanding of the ethical, social, political, and governance implications of artificial intelligence. It originates from a structured roundtable hosted by the Institute in July 2026 on the theme of AI and agriculture, with participants spanning philosophy, data science, rhetoric and design research, cybersecurity, physics, and agricultural work experience. After the roundtable, we verified claims against the public record and situated them within the scholarly literature on digital agriculture; sources are cited in the text and collected in the references. In keeping with Institute practice, ideas are presented collectively rather than attributed individually; the authors listed above are the roundtable's participants. The Institute intends this as the first in a series of papers examining artificial intelligence at its many intersections with domains of public consequence.

The method: interdisciplinarity as instrument

This paper's provenance is its method, and the method is the Institute's founding wager: that the questions artificial intelligence raises do not respect disciplinary borders, and so the thinking about them must not either. No single field could have produced this document. The applications catalogue draws on participants' hands-on agricultural and landscaping experience; the sovereignty analysis on a decade inside national research computing; the design critique on professional design research and rhetorical theory; the legal and policy threads on legal studies, cybersecurity practice, and data science; and the questions that bind them on philosophy, where the Institute began. The roundtable is the Institute's instrument, as the telescope is astronomy's: each meeting gathers disciplines around one domain, the conversation surfaces what no discipline sees alone, and papers such as this one are the recorded observations. Subsequent verification against the public record and the scholarly literature converts the conversation into a citable asset without flattening the plurality of expertise that produced it.

Executive summary

Agriculture has become one of the most consequential arenas of applied artificial intelligence, and one of the five priority sectors named in Canada's national AI strategy, released in June 2026 (Government of Canada, 2026). The applications are real and, in several cases, remarkable: laser-guided weeders that eliminate herbicide use, harvest robotics that pick only ripe fruit, satellite-driven precision planting, AI-assisted plant breeding, advisory tools that speak to farmers in their own vernacular, and controlled-environment systems that may relocate food production into cities and shipping containers. The benefits are measurable in water, pesticide, fertilizer, and cost savings.

Yet every promise arrives entangled with a question of power. Who can afford these tools, and what happens to the family farm that cannot? Who repairs a machine sealed by its manufacturer, and who owns the data harvested from a farmer's own field? What happens to soil, seed resilience, and rural livelihoods when optimization targets short-term yield? And what does sovereignty mean when a country's agricultural data, infrastructure identifiers, and AI systems are locked into platforms it does not control? The critical literature on digital agriculture has documented this pattern for a decade (Bronson and Knezevic, 2016; Rotz et al., 2019) and a landmark risk analysis in Nature Machine Intelligence formalized as systemic: risks of data interoperability and reliability, unintended socio-ecological consequences of yield-optimized models, and the safety and security of platforms deployed at scale (Tzachor et al., 2022). This paper maps both sides of that ledger and closes with recommendations and open questions that will shape the Institute's continuing work.

The promise: a field guide to applications

Participants began by cataloguing applications they had encountered firsthand in landscaping and farm work, as well as in research; each is corroborated by the public record.

Precision robotics

Laser weeding is the roundtable's emblematic case. Commercial systems now combine computer vision, deep learning, and industrial CO₂ lasers to identify weeds among crops in milliseconds and destroy them at the meristem with targeted thermal energy, eliminating herbicides, hand labour, and soil disturbance entirely; the leading platform reports weed-control cost reductions of up to eighty percent, payback periods of one to three years, capacity in the range of two hundred thousand weeds per hour, and deployment across fifteen countries (Carbon Robotics, 2022, 2026). Field research reports crop biomass increasing by more than thirty percent where laser weeding replaced herbicide applications (Carbon Robotics, 2026). Harvest robotics use sensor arrays to identify individual fruit at peak ripeness and pick selectively rather than stripping the plant, and a single autonomous machine can do the work of dozens of labourers almost continuously; some units amount to mobile data centres travelling the field.

Data, imagery, and breeding

Satellite imagery combined with machine learning identifies which crops need attention and maps efficient distribution of seed, fertilizer, and water. Adoption in Canada is already broad: by 2025, more than eighty percent of the country's largest farms and three-quarters of mid-sized operations were using or planning to adopt at least one digital agriculture tool, with machine-learning systems forecasting Ontario orchard yields months before harvest (Government of Canada, 2026). In plant science, models trained on troves of genetic and subspecies data surface breeding combinations promising higher yield or longer shelf life through conventional crossing rather than genetic editing.

Advisory intelligence and the vernacular problem

Large language model tools now offer farmers conversational advice on planting and management. Participants noted an equity concern the literature corroborates: farmers speak in local vernaculars, and advisory tools must be built with and for the communities they serve if access is to be genuinely equitable rather than nominally available. Tzachor et al. (2022) make the same point systemically, warning that expert systems designed without inclusive engagement will bypass precisely the smallholders and marginalized communities with the most to gain, and recommending participatory design that incorporates local and Indigenous knowledge systems.

Innovation from the Global South

Two applications presented at a recent agricultural summit convened by one of the participants illustrate a farmer-centric development model. In Barbados, an AI-powered monitoring tool aggregates photographs of leaves and plants submitted by farmers across the island to predict crop blights before they spread. A second tool curates databases of soil types alongside plant taxonomies and their common names across countries, allowing a newcomer to Canada to determine whether a traditional crop from Cambodia or Kenya can grow in a particular Canadian location. These projects treat farmers as data partners rather than data sources, and they point toward AI in service of food culture and migration rather than only yield.

Controlled environments and regenerative applications

With farmland prices climbing and weather growing more volatile, participants discussed the proposal of contained farming: controlled indoor structures, including shipping-container systems in which produce is grown in transit and arrives ready for harvest, with energy intensity as the standing trade-off. More speculatively, participants proposed AI-designed urban food forests, plant communities optimized to survive local climate with minimal maintenance while providing freely available food; AI-assisted soil

restoration through modelled crop rotation and placeholder planting, reviving practices that predate commercial agriculture; and applications to livestock health and sustainability, with knock-on potential for veterinary medicine.

The tensions

The environmental ledger

AI in agriculture sits on both sides of the environmental account. Training and running models consumes significant electricity, and the sector's growth is entangled with the data-centre expansion now contested in many municipalities. Yet deployed well, the same systems deliver substantial reductions in water, pesticide, fertilizer, and fuel use. The ledger does not balance itself; it must be balanced deliberately, with the full costs of computation weighed against measured savings in the field. Tzachor et al. (2022) propose one mechanism: staged, risk-aware deployment in supervised “digital sandboxes” where environmental and social impact assessment precedes scale.

Who can afford the future?

The family farm is challenged from two directions at once. Autonomous equipment reduces the need for labour in a sector already struggling to attract successors, and the capital cost of competitive AI equipment runs to hundreds of thousands of dollars. The critical literature documents exactly this dynamic: asset constraints produce inequities in access to infrastructure, equipment, and software that have impacted smaller farmers' ability to participate in digital agriculture at all, while the benefits of digitalization accrue disproportionately to consolidated operations (Rotz et al., 2019). Participants pressed the question plainly: can a family farm compete against operations that can afford the machines, and what happens to rural communities when it cannot?

Dependency by design: the right to repair

Farmers are, as one participant with farm experience put it, “a handy lot: when something breaks, they fix it.” Modern agricultural equipment is increasingly designed to prevent

exactly that, with diagnostic software and repair capability reserved to authorized dealers under warranty and digital-lock regimes; scholars have tracked farmers' organized advocacy for data ownership and the right to repair as a direct response (Rotz et al., 2019; Clapp and Ruder, 2020). The dependency has now been tested in law, twice in a single season. In April 2026, Deere and Company (commonly known as John Deere) agreed to pay ninety-nine million US dollars into a class-action settlement fund for farmers over repair restrictions, alongside ten years of access to the digital tools required for maintenance, diagnosis, and repair of large agricultural equipment (PIRG, 2026). On July 8, 2026, Deere settled the joint antitrust action brought by the United States Federal Trade Commission and five states, agreeing, under a decade of supervision, to provide farmers and independent repair providers the same repair resources and software capabilities available to its authorized dealers (Federal Trade Commission, 2026). Canada, notably, legislated first: following advocacy led by shortline manufacturer Honey Bee Manufacturing of Frontier, Saskatchewan, whose combine headers had been technically blocked from dominant platforms by digital locks, Parliament passed Bills C-244 and C-294, receiving royal assent in late 2024, permitting the circumvention of technological protection measures for the purposes of repair and interoperability respectively (Parliament of Canada, 2023; Miller Thomson, 2025). The roundtable's structural lesson stands on this record: repairability and interoperability must be secured in law and procurement before dependency hardens, not litigated afterward.

Whose data, whose sovereignty?

Farm data, on soils, yields, growing cycles, and inputs, is valuable no matter who owns the machinery, yet its governance remains largely a matter of the user agreements developers require farmers to sign, a fragmented and ambiguous arrangement the literature has flagged for years (Rotz et al., 2019; Clapp and Ruder, 2020). Canada's national *AI for All* strategy names agriculture among five priority sectors, dedicates a pillar to sovereign AI infrastructure, and

declares that data must be treated as "a strategic national asset" (Government of Canada, 2026); what it leaves unspecified is farm-level data governance, who owns the data harvested from a field, where it may be stored, and who may access it. The infrastructure layer deepens the concern.

Drawing on a decade of experience with national research computing, one participant described how the absence of sovereign, standardized identifiers for people, institutions, and data has already produced costly lock-in: AI-powered systems built on a major national research cluster face challenges in migrating to other platforms because authentication and identity are specific to that cluster. When a researcher changes institutions, they may lose access to their own intellectual property. This porting difficulty is significant given that the cluster is substantial: Nibi, a forty-three-million-dollar system at the University of Waterloo, commissioned in 2025 through the Digital Research Alliance of Canada and Compute Ontario. It supports thousands of researchers via the SHARCNET consortium (Digital Research Alliance of Canada, 2025; University of Waterloo, 2025). Commercial platforms compound the problem with deliberately non-standard authentication and connection layers. Participants pointed to a portability precedent from health informatics, in which patient data was architected as a function of the person, travelling with them across systems and borders, as the design philosophy agricultural data deserves: the farm's data as a function of the farmer.

The pace problem

A recurring theme: AI is being built inside an economic system that demands return on capital before all else, and it scales faster than law, custom, or communities can respond. Participants argued that policy and technology must move hand in hand, with governance present from the beginning rather than retrofitted after harms consolidate. Canada's recent history illustrates the difficulty: the Artificial Intelligence and Data Act, the country's first attempt at comprehensive AI legislation, lapsed in January 2025 without passage, and the

national strategy that replaced it in June 2026 favours adoption and investment while leaving binding obligations to future regulation (Baker McKenzie, 2026). Participants also noted that legal requirements to explain or understand model behaviour can outpace what current machine learning systems can show, particularly given differing views on what interpretability means. For this reason, the roundtable saw research that moves systems toward greater interpretability as an important support for governance, helping ensure that the law does not write cheques that technology cannot yet cash.

Designed in an ideological corner

Every product is an argument about how the world should work, and every system is constructed in some ideological corner. Applied to agriculture, the question is whether tools are farmer-centric or industry-centric: whether they are designed to scale and sell, or to serve the people who work the land. The scholarship is blunt about the failure mode: agricultural technology production is top-down, and developers frequently do not understand the domains they model (Rotz et al., 2019); responsible deployment requires structurally including farmers, local know-how, humanistic expertise, and Indigenous knowledge systems in design (Tzachor et al., 2022). Participatory design is not a courtesy in this domain; it is a requirement for accuracy as much as a moral obligation. Tools designed without the knowledge of those who work the land are likely to misrepresent the conditions they claim to model, so design processes must include multiple perspectives, focusing on the farmers, farmworkers, and Indigenous and local communities whose knowledge, labour, and livelihoods these systems will shape.

The ecology of optimization

Optimization has an ecology, and it can be destructive. Participants warned that over-engineered farming risks breeding out resilience: crops perpetually monitored and shielded from stress may lose the capacity to survive drought and disease, and models optimized narrowly for yield discourage crop rotation and soil care; the temptation to plant the money-making crop a

third season, because the price is high, is exactly the behaviour a naive objective function rewards. This is the risk class Tzachor et al. (2022) name unintended socio-ecological consequences of yield-optimized machine learning, and it sits within a longer scholarly debate over whether precision technologies serve environmental sustainability or entrench a technological lock-in antithetical to agroecological alternatives (Clapp & Ruder, 2020). What is optimized, over what horizon, for whose benefit: these are the real design decisions, and they are ethical before they are technical.

Openings: what better looks like

The roundtable did not end in critique. Participants sketched countervailing structures: decentralizing agricultural AI into smaller open-source components so that no single company can capitalize every layer of the industry, and so local manufacturers can serve local markets at prices family farms can bear; augmenting existing farm infrastructure, such as retrofitting intelligence onto irrigation already in the ground, rather than requiring wholesale replacement; public funding for public-interest agricultural AI, on the recognition that current incentives fund only what can be sold; genuine market competition to loosen incumbent control of the equipment layer, a loosening the 2026 settlements may accelerate; and development models, exemplified by the Caribbean tools described above, in which farmers participate as partners in the systems built from their knowledge. Several participants converged on a reframing: rather than replacing nature with machinery, use the machinery to help nature work better, from food forests to modelled crop rotation to soil restored by planting rather than chemistry.

Recommendations

From the roundtable's discussion, verified against the record and the literature, the Institute distils nine positions for policymakers, funders, and builders:

1. Secure repairability and interoperability in law and procurement before

- dependency hardens; the 2026 Deere settlements demonstrate the cost of the alternative, and Canada's Bills C-244 and C-294 offer a legislative template already in force.
2. Extend the national AI strategy's treatment of data as a strategic asset down to farm-level data governance: explicit rules on ownership, storage, and access for agricultural data.
 3. Establish sovereign, standardized identifiers and portability requirements across research and agricultural data infrastructure, so that systems and people are not locked to platforms.
 4. Treat the farm's data as a function of the farmer, adopting the portability architecture proven in health informatics.
 5. Fund interpretability research as a governance prerequisite, since law that requires understanding of models cannot function until such understanding exists.
 6. Mandate participatory, farmer-centric design in publicly funded agricultural AI, with domain knowledge, humanistic expertise, and Indigenous knowledge systems structurally included (Tzachor et al., 2022).
 7. Adopt staged, risk-aware deployment, piloting agricultural AI in supervised digital sandboxes with environmental and social impact assessment before scale (Tzachor et al., 2022).
 8. Support decentralized and open-source agricultural AI components as a matter of competition and food-system resilience.
 9. Require resilience accounting in agricultural optimization, so that soil health, crop diversity, and long-horizon sustainability are weighed alongside yield.

Questions for continued inquiry

The Institute carries forward the questions the roundtable opened but did not close. What would specifically Canadian applications of agricultural AI look like, and what national capacity do they require? How should the energy and water costs of agricultural computation be accounted against its field-level savings, and by whom? What governance would let controlled-environment agriculture grow without deepening energy dependency? Can urban food systems, from food forests to container farms, be designed as public infrastructure rather than commercial products? And beneath all of these: how do communities retain the power to decide what optimization is for?

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About the Ethics of AI Institute

The Ethics of AI Institute is an independent interdisciplinary organization serving as a forum for research, education, public engagement, and professional dialogue concerning the development, deployment, and governance of AI systems. Its mission is to promote informed and constructive engagement with the ethical challenges and opportunities presented by artificial intelligence through research, education, publications, events, and collaboration across academic, public, and private sectors.

Papers in this series are born of the Institute's roundtable meetings, which convene scholars, practitioners, students, and community members across disciplines. Readers who wish to join a future roundtable, propose a theme, or affiliate with the Institute are invited to write to the address below.

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